Tutorial No : 6

Tutorial Title: Implement Stack using Linked List.
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Program:

```
#include <stdio.h>
#include <stdlib.h>

struct node {
   int data;
   struct node *next;
};

struct node *top = NULL;

void push(int value) {
   struct node *newnode = (struct node*)malloc(sizeof(struct node));

   if (newnode == NULL) {
      printf("Stack overflow!\n");
      return;
   }

   newnode->data = value;
   newnode->next = top;
   top = newnode;

   printf("%d pushed into stack\n", value);
}

void pop() {
   if (top == NULL) {
      printf("Stack is empty\n");
      return;
   }

   struct node *temp = top;
   printf("Popped element = %d\n", top->data);
   top = top->next;
```

```
    free(temp);
}

void display() {
    struct node *temp = top;

    if (top == NULL) {
        printf("Stack is empty\n");
        return;
    }

    printf("Stack elements are:\n");
    while (temp != NULL) {
        printf("%d\n", temp->data);
        temp = temp->next;
    }
}

int main() {
    int choice, value;

    while (1) {
        printf("\n1. PUSH\n2. POP\n3. DISPLAY\n4. EXIT\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1:
                printf("Enter value: ");
                scanf("%d", &value);
                push(value);
                break;

            case 2:
                pop();
                break;

            case 3:
                display();
                break;

            case 4:
                return 0;
```

```
            default:
                printf("Invalid choice\n");
        }
    }
}
```

Output:

Output

```
1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 1
Enter value: 10
10 pushed into stack

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 1
Enter value: 5
5 pushed into stack

1. PUSH
2. POP
3. DISPLAY
4. EXIT
Enter your choice: 3
Stack elements are:
5
10
```